ReRAG: Retrieval-Augmented Generation for Reducing Hallucinations

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Abstract: Large Language Models (LLMs) frequently generate plausible yet ungrounded factual assertions, termed hallucinations. In this paper, we present ReRAG, an architecture designed for reducing factual hallucinations in retrieval-augmented generation. Standard single-hop RAG systems retrieve documents once and assume generation fidelity. In contrast, ReRAG incorporates an iterative audit loop that scores candidate assertions against retrieved evidence.

1. System Architecture

The primary architectural purpose of ReRAG is reducing factual hallucinations by validating claims against retrieved context passages. Candidate responses undergo sentence-level decomposition and lexical entailment scoring. If candidate faithfulness falls below the verification threshold, a refinement query is automatically synthesized, retrieving supplementary evidence from the underlying corpus.

2. Document Ingestion and Provenance Tracking

High-precision retrieval requires structured chunking. Text is partitioned into configurable overlapping chunks, preserving document provenance including source filenames and exact page numbers.

ReRAG Experimental Methodology & Results

3. Verification Threshold & Convergence

The verification step checks candidate claims against retrieved context passages using a consistency threshold of 0.7. If the candidate faithfulness score is greater than or equal to 0.7, the response is accepted. Otherwise, iterative refinement continues up to a maximum of 3 refinement steps. When evidence is insufficient, the model generates a constrained response acknowledging the limitation rather than fabricating facts.

4. Multi-Domain Scientific Evaluation

Benchmark experiments were conducted across scientific domains:

- Astronomy: The James Webb Space Telescope observes infrared wavelengths to detect early cosmic structures and distant primordial galaxies.
- Biology: Photosynthesis converts solar light into chemical sugars inside plant chloroplasts through light-dependent and Calvin cycle reactions.
- Quantum Computing: Quantum computing leverages superposition and entanglement to solve discrete optimization problems exponentially faster than classical algorithms.

5. Hyperparameter Sensitivity

Chunk size and overlap significantly impact context precision. Excessively small chunks truncate relational clauses, while overly large chunks dilute semantic density.